

Kamsani Yashwanth Chowdhary

ECE (VLSI) Undergraduate — FPGA • Edge AI • Embedded Systems
Hyderabad, Telangana, India — +91 72073 30433 — yashwanthchowdhary83@gmail.com
linkedin.com/in/kamsani-yashwanth-chowdhary — github.com/Yash1026-hash

TECHNICAL SKILLS

Hardware & RTL: Verilog HDL, SystemVerilog, FPGA (Zynq-7000), Vivado ML, AXI4-Stream, UVM, Cadence, LTSpice
AI / ML & DSP: YOLOv8, PyTorch, OpenCV, FIR/IIR Filters, FFT, DDS, CORDIC, MATLAB, NumPy
Programming: C, C++, Python, Bash, SQL
Embedded & Tools: Linux, ESP32, Raspberry Pi 4, SPI/I2C/UART, Docker, Git, VS Code

PROJECTS

- FPGA DSP Processing Chain** *Verilog, AXI4-Stream, Vivado 2024–2025*
- Built a 5-IP AXI4-Stream DSP pipeline integrating DDS, FIR, CIC, FFT, and CORDIC on Artix-7 FPGA using SystemVerilog wrappers, XDC constraints, simulation testbenches, and ILA-based hardware validation.
 - Implemented reusable parameterized clock-divider IP blocks and a 31-tap Hamming-window FIR low-pass filter using Vivado FIR Compiler IP for modular DSP integration.
 - Validated complete dataflow using waveform simulation, timing analysis, and hardware debugging with Integrated Logic Analyzer (ILA).
- Smart Urban Traffic Density Analyzer** *YOLOv8l, C++, Docker 2024*
- Developed a multi-threaded real-time traffic detection pipeline using OpenCV with Dockerized deployment for scalable and reproducible edge AI inference.
 - Optimized live video inference for urban traffic analytics and real-time congestion monitoring on edge-based computing systems.
 - Ranked **Top 3 among 200+ teams** at IISc Bangalore Urban Vision Hackathon.
- Thermal-Aware VLSI Floorplanning** *Python, MATLAB 2024–2025*
- Developed a Hybrid PSO + Simulated Annealing floorplanning prototype targeting HPWL minimization, congestion reduction, and thermal hotspot reduction using RC-grid thermal modeling.
 - Worked with MCNC benchmark circuits and collaborated in a four-member research team under faculty mentorship for academic documentation and evaluation.
- Autonomous Lane-Following Robot** *Python, OpenCV, Raspberry Pi 4 2023*
- Implemented real-time lane detection, contour extraction, and PID-based navigation on embedded Linux with 33 ms frame latency.
 - Integrated OpenCV-based computer vision with motor-control logic for autonomous steering and adaptive navigation.
 - Ranked **Top 3 among 150+ teams** at VR Siddhartha Engineering College Hackathon.

LEADERSHIP & EXPERIENCE

- Vice President — Aparmaya Club (AI & Autonomous Systems)** *Jan 2024 – Present*
- Lead 30+ student engineers building embedded and autonomous systems; reduced integration defects by 40% through structured code reviews.
 - Organized “Samyak” national-level technical fest for 200+ attendees and contributed to ECE Newsletter and PULSE broadcasting initiatives.
 - Coordinated technical events, mentorship sessions, and interdisciplinary project collaboration across AI and robotics domains.
- Treasurer — IEEE ComSoc Student Chapter** *Jan 2025 – Present*
- Managed chapter finances and operations for 5+ technical workshops and authored technical content reaching 1,000+ students.
 - Assisted in organizing seminars, technical outreach programs, and IEEE student engagement initiatives within the university.

EDUCATION

- KL Deemed University, Hyderabad** **May 2027**
B.Tech in Electronics and Communication Engineering (VLSI) CGPA: 8.29/10
- Anurag Engineering College, Telangana** **2024**
Diploma in Electronics and Communication Engineering CGPA: 9.18/10
- ZPHS Konijerla, Telangana** **2021**
Secondary School Certificate (SSC) CGPA: 10/10

ACHIEVEMENTS

- Top 3 — Urban Vision Hackathon (IISc Bangalore, 2024) among 200+ teams for AI-based traffic analytics.
- Top 3 — VR Siddhartha Engineering College Hackathon (2023) among 150+ teams for autonomous robotics systems.
- IEEE ComSoc Treasurer; led workshops and authored technical content reaching 1,000+ students.
- Vice President of Aparmaya Club leading 30+ student engineers in AI and autonomous systems projects.